

Java OOP Scenario-Based Daily Practice Assignment

Solve one problem every day. Each problem is designed to strengthen Object-Oriented Programming concepts such as Classes, Objects, Encapsulation, Constructors, Inheritance, Polymorphism, Abstraction, Interfaces, Collections, and Method Overriding. Create proper classes, attributes, constructors, getters/setters, and helper methods wherever applicable.

1. Smart Parking Management System

Design a Vehicle class containing vehicle number, owner name, and parking hours. Create a ParkingLot class to manage vehicles. Reject invalid vehicle numbers containing spaces and negative parking hours. Display all vehicles parked for 5 or more hours and calculate total parking revenue.

2. Library Book Management

Create a Book class and a Library class. Store book ID, title, borrower name, and days borrowed. Reject invalid book IDs. Display overdue books borrowed for more than 14 days and calculate overdue fines.

3. Employee Attendance Tracker

Create an Employee class with employee ID, name, and attendance percentage. Develop an AttendanceManager class that identifies employees below 75% attendance and generates attendance reports.

4. Movie Ticket Booking Portal

Create Customer and Booking classes. Store customer details and number of tickets booked. Prevent invalid bookings and identify customers who booked more than 5 tickets.

5. Hotel Reservation System

Design Guest and Reservation classes. Store guest details, room type, and bill amount. Find the guest with the highest bill and display a detailed reservation summary.

6. Online Course Learning Platform

Create Student and Course classes. Track completed courses and generate certificates for students who complete at least 3 courses.

7. Electricity Billing System

Create Consumer and BillCalculator classes. Store consumer details and units consumed. Calculate electricity bills using slab-based pricing and generate a bill summary.

8. Inventory Management System

Design Product and Inventory classes. Track stock quantities, product prices, and categories. Display out-of-stock products and inventory valuation.

9. Hospital Patient Queue Manager

Create Patient and QueueManager classes. Track patient details and waiting times. Display the patient waiting the longest and generate queue statistics.

10. Bank Account Management

Create an Account class with encapsulated balance fields. Implement deposit, withdraw, and balance inquiry methods. Identify accounts with balance below ■1000.

11. Bus Reservation Application

Create Passenger and Bus classes. Manage seat bookings and prevent duplicate seat allocations using proper validation methods.

12. College Result Processing System

Design Student and ResultProcessor classes. Store marks, calculate grades, identify toppers, and display failed students.

13. Restaurant Billing Software

Create MenuItem, Order, and Billing classes. Calculate table-wise bills and total daily revenue.

14. Mobile Recharge Management

Create Customer and Recharge classes. Store recharge history and identify customers with the highest recharge amount.

15. Warehouse Shipment Tracker

Design Shipment and Warehouse classes. Track shipment IDs and delivery days. Display delayed shipments and calculate average delivery time.

16. Gym Membership Management

Create Member and Membership classes. Track subscription plans and identify long-term members subscribed for more than 12 months.

17. Car Service Center Automation

Create Car and ServiceRecord classes. Track service history and calculate average service costs for all vehicles.

18. E-Commerce Order Processing

Design Product, Customer, and Order classes. Calculate order values, apply discounts, and identify premium orders above ■5000.

19. Flight Check-In System

Create Passenger and Flight classes. Track baggage weights and identify passengers carrying baggage above the allowed limit.

20. Hostel Room Allocation Manager

Design Student and Hostel classes. Allocate rooms, prevent duplicate allocations, and generate room occupancy reports.